

Yunhao Dong

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EDUCATION

New York University, Tandon School of Engineering

New York, NY

Master of Science in Computer Engineering

May 2027

Bachelor of Science in Computer Engineering | Dean's List | GPA: 3.87

Sep. 2022 - May 2026

TECHNICAL SKILLS

Languages: Python • C • C++ • SQL • Swift • C# • Assembly • Verilog • Bash/Shell

Embedded: ARM Cortex-M • RTOS • Zephyr • SPI • BLE • I2C • UART • Linux • ISR • GPIO • DMA • DSP

Tools: Git • IAR • JTAG/SWD • Wireshark • Altium • MATLAB • Logic Analyzer • Oscilloscope • AWS EC2

EXPERIENCE

Mettler-Toledo Rainin

June 2026 – Aug. 2026

Embedded Software Engineer Intern

Oakland, CA

- Deployed pipette leak detection firmware for an STM32 system, achieving 98% accuracy and a 75% reduction in model memory, by deploying an Edge AI inference pipeline optimized via quantization and feature engineering
- Upgraded legacy firmware to support closed-loop neural network inference, cutting execution latency to under 100ms with multithreaded RTOS task scheduling in a state machine, utilizing queues, mutexes, and semaphores
- Automated data collection by python scripting to control Universal Robot arm, reduced test time by 40%

NYU Tandon Computer Science & Engineering Department

Sep. 2024 – May 2026

Course Assistant

Brooklyn, NY

- Conducted office hours weekly to assist 100+ students to clarify concepts, course material, and coding assignments
- Facilitated lab sessions on Python to guide 30 students through hands-on exercises to ensure topic comprehension

Honeywell International, Building Automation Team

June 2025 – Aug. 2025

Application Engineer Intern

Northford, CT

- Implemented firmware test infrastructure for BLE Channel Sounding using Zephyr RTOS on Nordic and Silicon Labs platforms, validating indoor positioning algorithms achieving 80% localization accuracy
- Designed and prototyped a low-power BLE beacon firmware for asset tracking, analyzing peripheral power states and duty cycles to optimize battery life through low-power modes and sleep/wake states

NYU Vertically Integrated Projects: Robotics Design Team

Jan. 2024 – May 2025

Systems Engineer

Brooklyn, NY

- Created firmware for manual & autonomous motor control via CAN bus in NASA's Lunabotics Competition
- Developed and optimized I2C driver communication on embedded Linux (NVIDIA Jetson Orin Nano) with Teensy
- Debugged signal integrity issues across PCB for hardware bring-up, and wrote low-level drivers for sensors

NYU Undergraduate Summer Research

June 2023 – Aug. 2023

Student Researcher

Brooklyn, NY

- Integrated UDP/TCP networking & computer vision to track hand movements in Unity with <50ms latency
- Designed embedded hardware PCB in Altium for sensor data acquisition, reducing prototype footprint by 40%

PROJECTS

Kira | Edge Impulse, Nano 33 BLE, Whisper, Google Calendar

May 2026 – July 2026

- Built a voice-controlled assistant integrating UART, Whisper speech-to-text, & Google Calendar API
- Implemented MFCC audio feature extraction pipeline in C++ (FFT, mel filterbank, DCT) for real-time signal processing and wake-word detection under strict memory constraints on a bare-metal microcontroller

Postura | Arduino, iOS, ML, Python, TensorFlow Lite

Oct. 2025 – May 2026

- Developed and deployed a neural network posture classifier on an Arduino Nano 33 BLE with TensorFlow Lite, achieving 95%+ accuracy within flash and RAM constraints
- Engineered a custom iOS app integrating BLE communication interfaces to stream live posture classifications

RISC-V Simulator | RISC-V, C++, Pipelined ISA, Algorithm Development

Jan. 2025 – May 2025

- Architected a cycle-accurate RISC-V ISA simulator in C++, including instruction fetch/decode, register file state management, and memory-mapped access
- Integrated hazard detection and pipeline control logic, including instruction stalling and data forwarding